Student's Index Number ……

THE PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT AUTHORITIES

ARUSHA REGIONAL MOCK EXAMINATION

FORM TWO

031 **PHYSICS**

TIME 2:30HRS **MAY 2026**

ARUSHA REGIONAL MOCK EXAMINATION

INSTRUCTIONS

1. This paper consists of three sections A, B and C with a total of 10 questions
2. Answer all questions in section A, B and C
3. Read carefully the instructions given in each section
4. Write your examination number on every page
5. Where necessary the following constant may be used
   (i) Acceleration due to gravity = 10N/kg or $10m/s^2$
   (ii) Density of water = $1g/cm^3$=$1000Kg/m^3$
   (iii) Resistivity of constantan wire ρ = 49 x 10-7 Ωm.
   π=3.14

QUESTION NUMBER	SCORE	EXAMINER'S SIGN
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
TOTAL		
CHECKER' SIGN		

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SECTION A (15 Marks)

1. For each of the items (i) – (x) choose the correct answer from among the given alternatives and write its letter besides the box provided.

(i) When a large body of experimental evidence supports the hypothesis or does not support the hypothesis, What may the hypothesis eventually be considered?

A. An observation
B. An insight
C. A conclusion
D. A law

(ii) Which of the following best describes a variable?

A. A trend that shows any exponential relationship
B. Something whose value can change over multiple measurement
C. A measure of how much a plot line changes along the y – axis
D. Something that remains constant over multiple measurement.

(iii) To measure the length using a meter rule, wrong position of the eye leads to

A. A parallax error
B. Eye error
C. Zero error
D. Metre error

(iv) A certain student use to push a wall using his hands but he found no any movement on the wall. As a form two student comment on this

A. Work is done by the student on the wall
B. Work is done by the wall on the student
C. No work is done at All
D. Some of work is done on the wall

(v) Which of the following instrument is most suitable for measuring the internal diameter of 100ml beaker?

A. Metre rule
B. Vernier caliper
C. Measuring tape
D. External caliper

(vi) The mass of an object is reduced if its,

A. States its change
B. Amount of matter is reduced
C. Surface is reduced
D. Volume is reduced

(vii) Which of the following results is the correct mass of the glass that has the same volume as 5.4g of aluminum?

A. 2kg B. 5.0g C. 0.5g D. 2.5g

(viii) A property of a body to behave like elastic membrane

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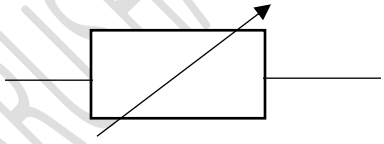
A. Surface tension
B. Capillarity
C. Cohesion
D. Adhesion

(ix) If the relative density of a substance is 11.3, then its density is
A. 11.3kg/m^3
B. 11.3g/cm^3
C. 0.0113g/cm^3
D. 11300g/cm^3

(x) One of the following is pair of forces
A. Gravity and diffusion
B. Attractive and repulsive
C. Up thrust and gravity
D. Osmosis and diffusion

i	ii	Iii	iv	v	vi	vii	viii	ix	x

2.Match the items in List A with response in List B by writing the letter of the correct response beside the item number in the space provided.

LIST A	LIST B
i 	A. Battery
ii 	B. Conductor (wire)
iii 	C. Fixed resistor
iv 	D. Rheostat
v	E. Capacitance
	F. Capacitor
	G. Cell
	H. Thermistor

Answers

i	Ii	iii	iv	v

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SECTION B(70 Marks)

3.(a) (i) Differentiate magnetization from demagnetization

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(ii) Outline any three properties of magnet

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(b)(i)Briefly explain three ways of storing magnets.

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(ii)As you discussed the concepts of magnets, describe four uses of magnets in daily life

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4. (a) (i) Differentiate between density and relative density

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(ii)Outline three applications of density in real life.

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(b)(i)List down three factors affecting density

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(ii)A stone has a mass of 112.5g. When the stone was totally immersed in water contained in measuring cylinder displaced water from $50cm^3$ to $95cm^3$.

(iii)Continue find the density of the stone

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5. (a)(i) State Archimedes principle

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(ii) An iron piece of mass 360g and density of $7.8g/cm^3$ is suspended by a rope so that it is partially submerged in oil of density $0.9g/cm^3$. Find the tension in the string.

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6. (a)(i) State the principle of conservation of

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(ii)Explain briefly the energy changes using a simple pendulum bob as it swings from one side to another.

(b) Define the following terms

(i) Potential Energy

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(ii)Kinetic Energy

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7. (a)(i)Explain why water wets glass but mercury does not

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(ii)Describe the factors affecting capillarity

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(b)(i)Briefly explain why a potter put some soft material on his/her head for a heavy load

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(ii)What are the characteristic of pressure in liquid?

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8. (a) Define the following terms as applied in curved mirror

(i) Radius of curvature

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(ii) Focal length

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(b)(i)State four rules used to locate an image formed by a concave mirror

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(ii)An object 20cm high is placed 40cm in front of a concave mirror of focal length 15cm. Determine the position, nature, size and magnification of the image formed using a ray diagram.

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9. (a) Define the following terms

(i) Uniform velocity

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(ii) Retardation

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(b)A train accelerates uniformly from rest to a speed of 80m/s in 20seconds. It maintained this speed for 15 seconds and finally decelerates uniformly by applying a break and comes to stop after 25 seconds.

(i) Draw a graph to show the whole Journey

(ii) Calculate the total distance covered by a train

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(iii) Calculate the acceleration of the motion

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SECTION C (15 MARKS)

10. (a) (i) What is a Gold leaf electroscope

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(ii)Draw a well labelled sketched diagram of the device defined above

(b)(i) By using the concept of electrostatics explain why a person walking on the carpet floor/feel an electric shock when touches a metallic door knob

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(ii)If you are given two metal spheres standing on insulator stands and a positively changed rod, explain how you can change the metal with equal and opposite changes by inductions.

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(c) (i) State Ohm's law

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(ii)Outline four factors that affects resistance of a conductor.

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(iv) A constantan wire has a length of 45cm, and a diameter of 0.37mm
What is the resistance of the wire

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